

Computing Methods for Physics 1 – 11 September 2024

Your exam material (code files, plots, datafiles, etc.) must be submitted via google classroom by 12:45 as a single zip file.

C++ evaluation will be based on: correct syntax, proper return types, proper arguments of functions, data members and class interfaces, setters/getters, unnecessary void functions, correct mathematical expressions, comments throughout the code, separation of class implementations and interfaces.

Python evaluation will be based on: correct syntax, avoiding C-style loops, using Python features in general, comments throughout the notebook/scripts, labels, legends and plot styling, clarity in general.

Part 1 – C++

Write a class **Frac** to represent fractions. The class will require at least two members to represent the numerator (**n_**) and the denominator (**d_**). Provide the class with the following constructors:

1. a regular constructor;
2. a default constructor that interacts with the user asking for the values of numerator and denominator;
3. a copy constructor;
4. a constructor that assumes that the denominator is 1 and only takes the numerator as an argument.

Provide the class **Frac** with:

- getters and setters;
- a method to return the inverse of the fraction;
- a method to simplify the fraction.

Finally, overload the operators **+**, **-**, *****, **/**, and **<<** in the **Frac** class.

N.B. – Think of how you want to handle the sign of a fraction representation with a **Frac** instance: via the sign of **n_** keeping **d_** positive, or vice versa, or with a specific data member.

Part 2 – Python

Again write a class to represent fractions, this time in Python. You may use scripts or a notebook, as you wish. The tasks are similar to the ones for C++:

1. write the constructor so that the denominator is 1 by default;
2. provide getters and setters;
3. implement a method to return the inverse of the fraction;
4. implement a method to simplify the fraction;
5. overload the operators `+`, `-`, `*`, and `/`;
6. override the built in `str` function so that the user can `print` an instance of `Frac`.